

Program 3:

Code a Dockerized Python Flask or Node.js application

07/11/2025

Project Directory:

- |-requirements.txt
- |-app.py
- |-Dockerfile

Program code and steps for execution:

Step 1: Create a directory with the prog3 name in order to execute the program

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~$ cd mylab/  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab$ mkdir prog3  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab$ cd prog3/
```

Step 2: Now create the file called requirements.txt app.py and Dockerfile

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ touch requirements.txt Dockerfile app.py  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ ls  
app.py Dockerfile requirements.txt
```

Step 3: The requirements.txt

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ sudo nano requirements.txt  
[sudo] password for 1RV24MC030_chinmayi_m_h:  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ cat requirements.txt  
Flask==2.3.3
```

Step 4: The Dockerfile

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ sudo nano Dockerfile  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ cat Dockerfile  
# Use official python image  
FROM python:latest  
  
# Set the working directory  
WORKDIR /app  
  
# Copy the requirements.txt and install the dependencies  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
  
# Copy the application/source code  
COPY . .  
  
# Expose the port to run  
EXPOSE 5000  
  
# Run the app  
CMD ["python", "app.py"]
```

Step 5: The app.py

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ sudo nano app.py  
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ cat app.py  
from flask import Flask  
  
app=Flask(__name__)  
  
@app.route("/")  
def home():  
    return "Hello this is the simple Flask Docker, Welcome you all !!!"  
  
if __name__=="__main__":  
    app.run(host="0.0.0.0", port=5000)
```

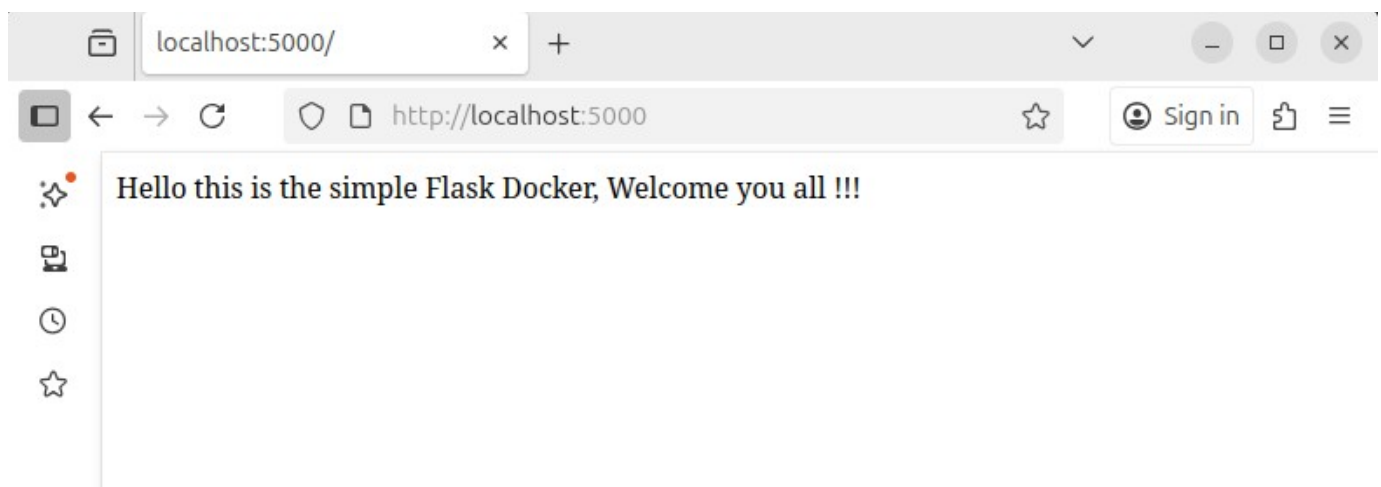
Step 6: Now run the flask application → `docker build -t <project_name> .`

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ docker build -t prog3 .
[+] Building 2.7s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 385B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28028afd858185d73bf4d6a439cbb058b6800a96d
=> [internal] load build context
=> => transferring context: 685B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:64352e7d9d753ce6b8a21e60f6ea3f00cee7bcdefc0ebcdebfd1397da8cae006
=> => naming to docker.io/library/prog3
```

Step 7: Now run the container → `docker run -p 5000:5000 <project_name>`

```
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ docker run -p 5000:5000 prog3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
```

Step 8: Now open the browser and type <http://localhost:5000/>



Step 9: Check for the images and containers then stop and remove the container also remove the images

```
^C(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
prog3 latest 64352e7d9d75 About a minute ago 1.13GB
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
3e4d805cca71 prog3 "python app.py" About a minute ago Exited (0) 8 seconds ago eloquent_black
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ sudo docker container stop 3e4d
3e4d
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ sudo docker container rm 3e4d
3e4d
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ sudo docker rmi 6435
Untagged: prog3:latest
Deleted: sha256:64352e7d9d753ce6b8a21e60f6ea3f00cee7bcdefc0ebcdebfd1397da8cae006
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
(base) 1RV24MC030_chinmayi_m_h@chinmayi-m-h-HP-Pavilion-Laptop-15-eg3xxx:~/mylab/prog3$
```